Among two-row tableaux the odd descent set determines the tableau uniquely

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Editorial note. The counted statement is the author's labelled lemma 'lem:superstandard two rows characterize coefficients' (source lines 594-601); the proof environment begins after blank lines, so the pairing is by adjacency.

Editorial note. The retained context is the author's own lead-in sentence announcing the stronger two-row property.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source; packages needed only by other parts of the article are left out because no retained line uses them.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: uniqueness of two-row standard Young tableaux with odd descent set (author label lem:superstandard two rows characterize coefficients).

Allowed standard prerequisites (editorial).

- Two-row standard Young tableaux. For lambda = (N-k_1,k_1) the superstandard tableau T_lambda fills the first row with the odd numbers and the second row with the even numbers up to 2k_1, and the rest in the first row.. The proof identifies row_2(T) with the even numbers {2,4,...,2k_1} using precisely this filling.
- Descent set and odd numbers. Des(T) is the set of i with i+1 in a lower row than i, and odds(k_1) = {1,3,...,2k_1-1}.. The statement is Des(T_lambda) = odds(k_1), and the proof reads the row membership off from that equality.

Standard Young tableaux with at most 2 rows have a slightly stronger property:

Lemma 0.1. Let $\lambda = (N-k_1, k_1)$ and $\mu = (N-k_2, k_2)$ be two partitions of N with at most two parts each. Then:

1. $\mathrm{Des}(T_\lambda) = \mathrm{odds}(k_1)$.
2. If $\mathrm{Des}(T) = \mathrm{odds}(k_1)$ for some $T \in \mathrm{SYT}(\mu)$, then $k_1 = k_2$ and $T = T_\lambda$.

Proof. The first assertion is obvious, so let us focus on the second assertion.

Let $T \in \mathrm{SYT}(\mu)$ be a tableau, and suppose that $\mathrm{Des}(T) = \mathrm{Des}(T_\lambda)$. Given that both T and T_λ have two rows each, it suffices to show that $\mathrm{row}_2(T) = \mathrm{row}_2(T_\lambda)$. Since $\mathrm{Des}(T) = \{1, 3, \ldots, 2k_1 - 1\}$, it follows that $\{1, 3, \ldots, 2k_1 - 1\} \subseteq \mathrm{row}_1(T)$ and $\{2, 4, \ldots, 2k_1\} \subseteq \mathrm{row}_2(T)$. Consequently, we have $2k_1 + 1 \in \mathrm{row}_1(T)$. As T has no descents beyond index $2k_1 - 1$, all entries greater than $2k_1$ must appear in the first row. Therefore, we conclude that $\mathrm{row}_2(T) = \{2, 4, \ldots, 2k_1\}$. □